open to internship, jan to jun 2027

Rizki Ramadhani

I build ba

Computer Engineering Technology student at IPB University, GPA 3.77 of 4.00. I take a signal from the sensor to the screen: firmware and PCB design, backend APIs, ML forecasting, and dashboards.

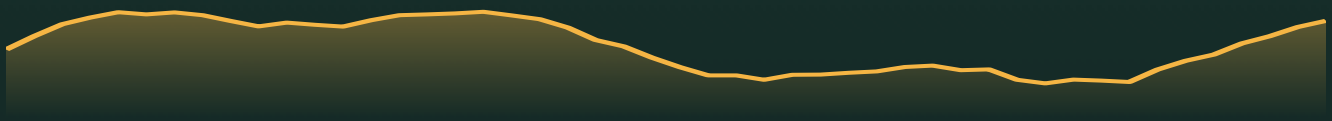
See the MUSH system

Email me

mush / drum-01

simulated signal

34.7 °C



gas index	humidity	pressure
479	65%	1010 hPa

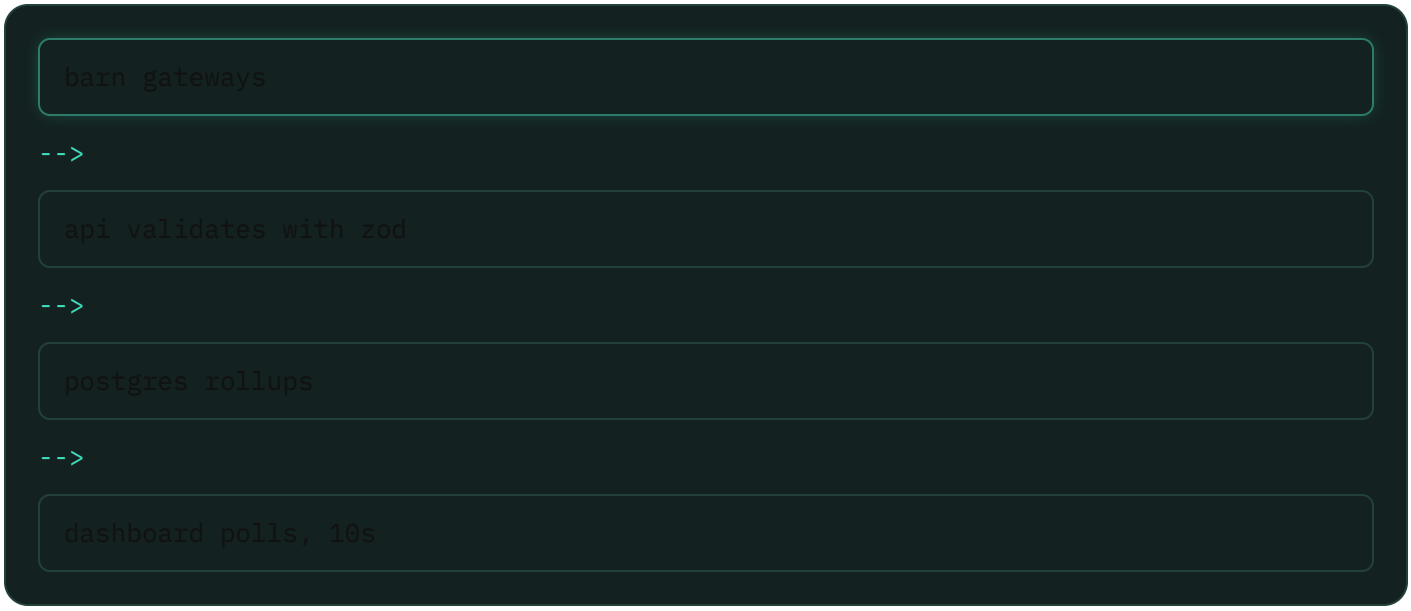
Demo signal in the style of the MUSH dashboard. Not live farm data.

public repos	8
agritechjam 2025	1st place
ipb gpa	3.77 / 4.00

flagship, 2024 to present, 4 repos

MUSH watches silage ferment at Pagi Farm

Silage fermentation happens inside sealed drums, invisible until the batch spoils. MUSH fixes that: ESP32 gateways report DS18B20 temperature, humidity, pressure, MQ-135 gas, and load-cell weight. One API stores it all, a sidecar service forecasts it, and a dashboard plots it.



MUSH Server, the central API

adonisjs, typescript, prisma, postgresql, docker, zod

- I built the REST API that ingests telemetry from ESP32 gateways and validates every payload with Zod.
- It rolls readings into hourly and daily aggregations, and provisions a drum record automatically when an unknown device ID shows up.
- When the ML service is down, the API answers 503 instead of failing silently.

github.com/rizkiramadhani-001/mush-server

MUSH ML, forecasting sidecar

`python, fastapi, scikit-learn, pydantic, pytest`

- I built a decoupled FastAPI microservice that trains and serves scikit-learn forecasting models on multisensor fermentation data.
- Pydantic guards the inputs and Pytest covers the training path.

github.com/rizkiramadhani-001/mush-ml

MUSH React, dashboard

`react, ionic, vite, three.js, recharts`

- I built a responsive web and mobile dashboard with an interactive Three.js 3D drum status view and Recharts time-series graphs.
- It ships light and dark themes with 10-second background polling, plus mock and HTTP data adapters.

github.com/rizkiramadhani-001/mush-react

MUSH landing page

`html, tailwind css, vanilla javascript`

- A static product page that presents the MUSH project to farm visitors, built with HTML, Tailwind CSS, and vanilla JavaScript.

github.com/rizkiramadhani-001/landing

`selected work`

Index

nov 2025

1st place

UMKM investment and collaboration platform

The problem: UMKM financial data is sensitive, yet investors need to see it before committing. The platform solves it with strict role separation.

- As backend developer I designed the Laravel and PostgreSQL backend for six roles (public, UMKM owner, investor, supplier, distributor, super admin), with access control that keeps each UMKM's financial data visible only to approved investors.
- I built the endpoints behind the real-time financial dashboards, revenue-based profit-sharing calculations, and automated monthly reports exportable to PDF and Excel.
- I integrated OpenSign e-signatures for investment and supply contracts, Pusher real-time chat, and a Gemini chatbot for FAQs and UMKM recommendations.

laravel, postgresql, react, tailwind, pusher, opensign, gemini api
team kabel hdmi, himalkom ipb, agritechjam 2025

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hardware

Multi-node RF433 telemetry and PCB hardware

The problem: raw sensor signals drift and pick up noise across distributed nodes. Stable readings need conditioning in hardware, not just filtering in code.

- I programmed C++ firmware for 4 LGT8F32P Nano-class sensor nodes and an Arduino Mega master board to acquire multi-point sensor data and transmit it wirelessly over RF433 modules.
- I designed custom analog sensor-conditioning schematics and PCB layouts in KiCad to stabilize signal acquisition across the distributed nodes.

github.com/rizkiramadhani-001/Arduino-Project-4-Node-and-Mega

c++, arduino mega, lgt8f32p, rf433, kicad

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web app

MyKeuangan, personal finance and reporting

The problem: personal budgets live in scattered notes, so monthly spending stays invisible. MyKeuangan puts records, categories, and budgets in one reactive app.

- I built a reactive finance app with Laravel and Livewire to record transactions, manage expense categories, and track monthly budgets.
- Filtered transaction reports export to Excel and PDF.

github.com/rizkiramadhani-001/MyKeuangan

laravel, php, livewire, tailwind, mysql

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more from github, all 8 repos

The rest of the shelf

sensorix-porto

css, forked from nabeelarrafi95-cmd/sensorix-porto

Portfolio site for the Sensorix team, kept as a fork I contribute to and learn from.

github.com/rizkiramadhani-001/sensorix-porto

rizkiramadhani-001, profile readme

profile, featured project mush

My GitHub profile page: featured project MUSH with its four repos, plus internship interests in full-stack web, backend systems, and IoT product development.

github.com/rizkiramadhani-001/rizkiramadhani-001

toolbox

Stack

languages

typescript javascript python php c c++ sql

backend and ml

adonisjs laravel livewire fastapi node.js prisma zod pydantic scikit-learn gemini api

iot and embedded

esp32 arduino mega lgt8f32p rf433 telemetry c/c++ firmware signal conditioning kicad pcb

frontend, data, tools

react ionic vite three.js recharts tailwind postgresql mysql docker git pytest linux

school and contests

About

Education

IPB University, College of Vocational Studies, Bogor, Indonesia. Applied Bachelor (D4) in Computer Engineering Technology, Aug 2024 to expected 2028. Cumulative GPA 3.77 of 4.00.

coursework

Embedded Systems, Microcontroller Architecture, Computer Networks, Database Systems, Object-Oriented Programming, Data Structures and Algorithms.

Timeline

2024	Started D4 Computer Engineering at IPB, began MUSH for Pagi Farm
2025	1st place, AgriTechJam 2025, backend developer, team Kabel HDMI
2026	COMPFEST 18 AI Innovation Challenge and Garuda Hacks 7.0 participant
2027	Open to 6-month credit-bearing internship, Jan to Jun

contact

Hire me for Jan to Jun 2027

A 6-month credit-bearing internship in backend, IoT, or embedded engineering through IPB University. Email is fastest.

Email Rizki | Copy email | Download CV

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